

Assignment 1

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1 Problem 1

Predict: the first four moments, and which distributions they rule out

The first four sample moments of x ($n = 1000$) are:

Moment	Value
Mean	0.001104
Variance	0.0000963
Std..	0.009811
Skewness (3rd moment)	-0.6688
Excess Kurtosis (4th moment)	2.3398

(a) Week 1, Section 6 (“Choosing a Distribution from the Sample Shape”) gives a filter based on the sign of the skew and the sign of the excess kurtosis. Our sample has skewness clearly different from zero (-0.669) and a large, clearly positive kurtosis (2.34). The table in that section says a sample with skew $\neq 0$ and excess kurtosis > 0 indicates Normal Inverse Gaussian.

- **Normal** is ruled out by both moments: a Normal has skew $= 0$ and excess kurtosis $= 0$ by definition (Week 1, Table 2), and our sample misses both targets.
- **Symmetric Student’s t** is ruled out by the skewness moment. its theoretical skew is always 0 (Week 1, Table 4). Since the skew is -0.67 , it cannot be explained.
- **Lognormal** is ruled out by both the support and the skewness moment. A Lognormal only has support on $(0, \infty)$ (Week 1, Table 3), but 406 of the 1000 observations in this sample are negative. Also, a Lognormal has strictly positive skew for every $\sigma > 0$, and our sample skew is negative.
- **Candidates that survive:** a skewed family with fat tails. The Normal Inverse Gaussian (Week 1, Section 5.5) or a skewed generalization of the t can make it, since these are the only families in the Week 1 that can carry a non-zero skew together with positive excess kurtosis.

Fit and reconcile

Matching the mean and variance gives $\mu = 0.001104$ and $\sigma = 0.009811$.

(b) The fitted Normal’s 1% quantile is $q(0.01) = -0.021719$. Out of 1000 observations, **26 fall below this value**. Under the Normal model, we should expect exactly 1% of 1000 observations, which is: **10 observations**, to fall below the model’s own 1% quantile by construction. The sample delivers **2.6 times** as many left-tail exceedances as the fitted Normal predicts.

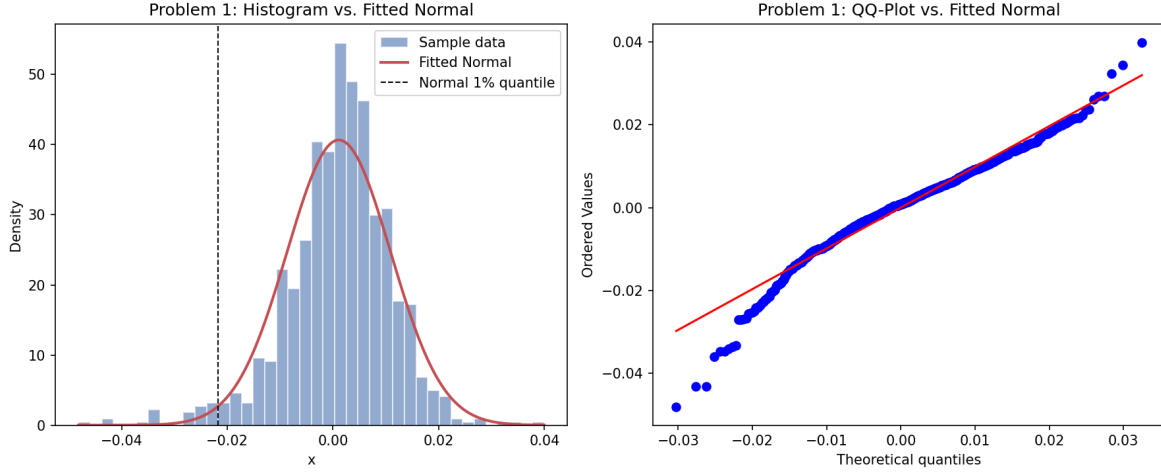


Figure 1: Left: histogram of x with the fitted Normal. Right: Normal QQ-plot of x , showing the systematic curvature in both tails

(c) If we used this fitted Normal as the model for the series, it would get the tails wrong, and it would get them wrong in a specific, asymmetric direction. The error is concentrated in the **left tail**: the Normal understates the probability of a large negative outcome by more than a factor of two (26 observed vs. 10 expected below the 1% quantile). This is exactly what a risk manager cares most about, since VaR and other loss measures live in that same left tail (Week 1, Section 2.3). The right tail tells the opposite story, that only 6 observations exceed the fitted Normal's 99% quantile (0.023927) against 10 expected. So probability mass that a symmetric model would place in the right tail has, in the real sample, effectively been pulled over into the left tail. That reallocation of tail mass from right to left is precisely what a negative skewness coefficient means, and it is invisible to a Normal fit because a Normal only ever sees the mean and variance, never the asymmetry.

2 Problem 2

Predict

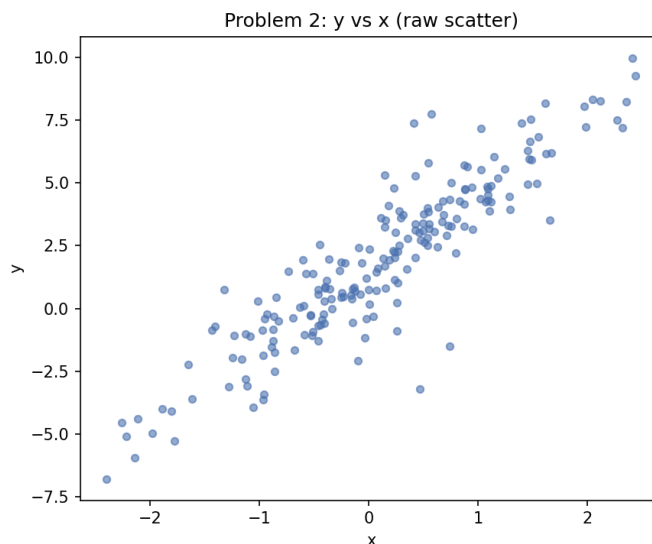


Figure 2: Raw scatter of y against x .

(a) The scatter shows a strong positive linear trend, but a few points (near $x \approx 0.5, y \approx -3.2$ and $x \approx 0.75, y \approx -1.5$) sit far off the line relative to how tight the rest of the cloud is. That pattern shows a tight core with occasional large excursions, which points to a **fat-tailed error**, so I expect a **Student's t** error (Week 1, Section 5.4) rather than a Normal one.

(b) This violates **assumption 7** of OLS, that the error is normally distributed (Week 2, Section 3.1). For Assumptions 1–6, which give $\hat{\beta} = \text{cov}(x, y) / \text{var}(x)$ (Week 2, Section 3.2), never use normality, so the **slope estimate moves very little**, but the **reported OLS standard error, and any exact-normal-theory inference built on it, becomes unreliable**.

Fit

Model	α	β	Error parameters	Log-likelihood.
1. OLS	1.5607	2.9908	$\hat{\sigma} = 1.3446$	−343.01
2. MLE, Normal	1.5607	2.9908	$\hat{\sigma} = 1.3446$	−343.01
3. MLE, Student's t	1.5480	3.0192	scale = 0.9355, $\nu = 3.632$	−328.56

Models 1 and 2 are numerically identical, since OLS is the Normal MLE (Week 2, Section 4.3). With $k = p + d$ ($p = 2$, $d = 1$ for Normal, $d = 2$ for t ; Week 2, Section 5.3):

$\Delta \text{AICc} = \mathbf{26.82}$. It shows that **AICc strongly prefers the t model**.

Model	k	AICc
OLS / MLE Normal	3	692.14
MLE Student's t	4	665.33

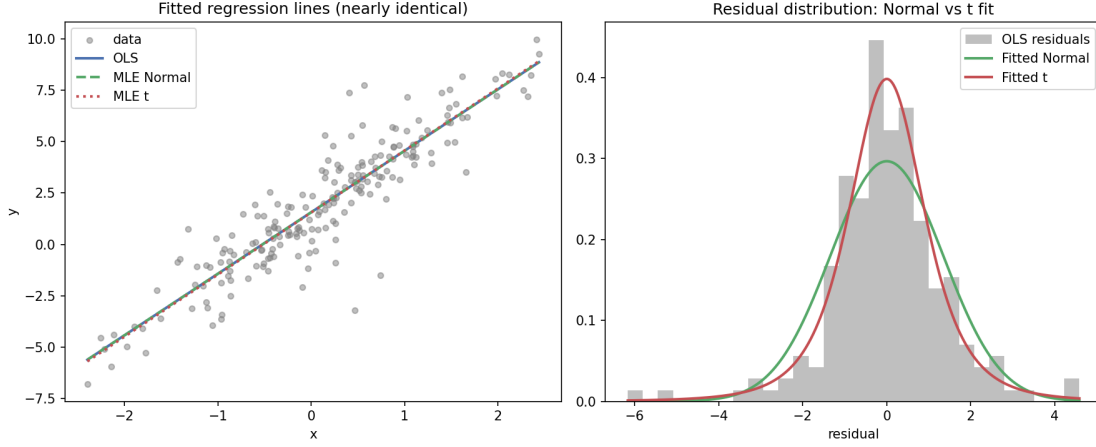


Figure 3: Left: the three fitted lines nearly coincide. Right: residuals with fitted Normal and t densities overlaid.

Reconcile

(c) The three slope estimates are 2.9908 (OLS), 2.9908 (Normal MLE), and 3.0192 (t MLE). They have a difference of about 0.028, roughly 1% of the slope itself, and well inside one OLS standard error (0.0966). This matches the prediction in part (b) almost exactly: **the violation barely moved the slope at all**. Non-normal errors, on their own, are not a source of bias in the slope; they change how much we should trust the reported precision of that slope, not the slope's central estimate.

(d) I think the Normal model is not wrong about the slope, it is wrong about **the tails**. Forced to fit the occasional large residual with a single σ , it ends up too wide for typical residuals and still too narrow for the extreme ones. The t model instead uses a smaller scale plus heavy tails to fit both at once. the AICc gap is about tail shape, not the central fit.

(e) Quantiles:

Quantile	Normal	t
95.0%	2.2117	2.0538
99.5%	3.4635	4.6201

The Normal is wider at 95%; the t is wider at 99.5%. The t is more sharply peaked, so its moderate quantiles can be smaller, but its heavy tail ($\nu \approx 3.6$) overtakes the Normal deep in the tail. For a **capital buffer**, which protects against extreme losses, the t 's 99.5% quantile is the right choice, and the normal would understate it by about 33%.

3 Problem 3

Predict

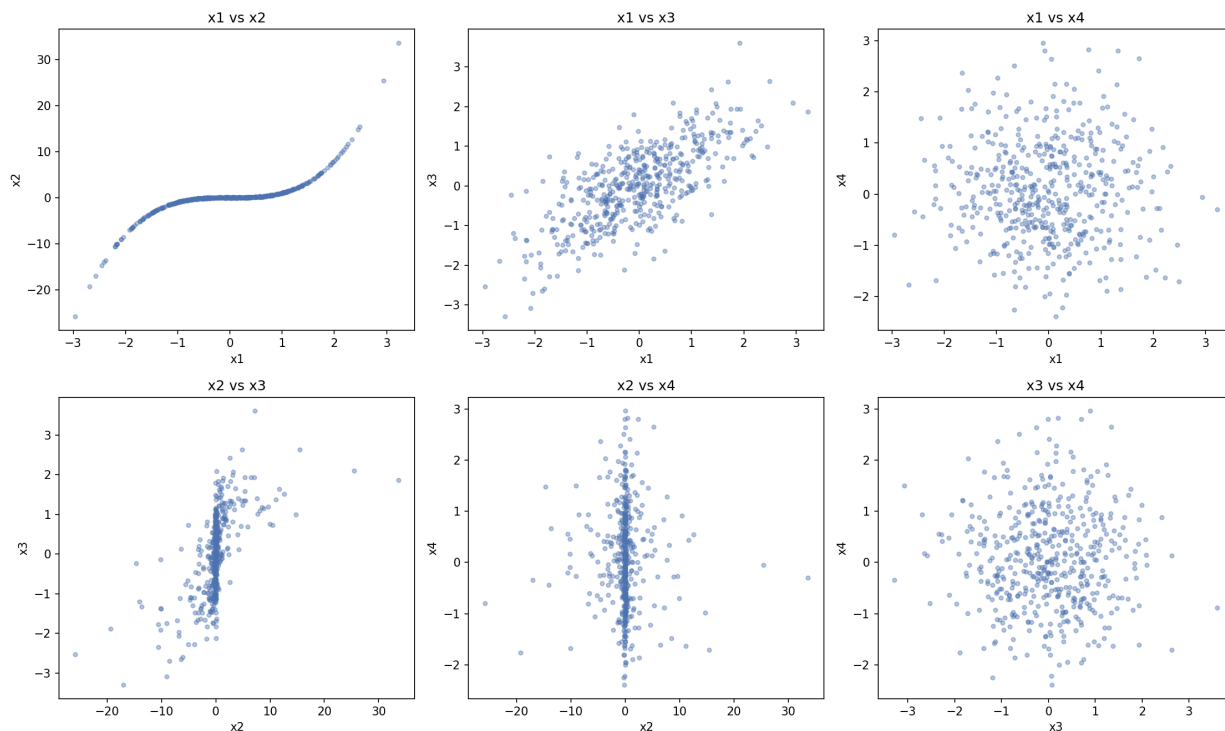


Figure 4: All six pairwise scatter plots among x_1, x_2, x_3, x_4 .

(a) x_1 vs. x_2 is the pair to watch: a tight, **curved** (cubic-like) relationship. Since Pearson only measures linear association (Week 2, Table 1), we expect x_1 vs. x_2 to show the **largest Pearson/Spearman gap**. x_1 vs. x_3 and x_2 vs. x_3 look like ordinary noisy, roughly linear clouds, so we expect the two measures to agree there. x_4 shows no relationship with anything, so both measures should agree near zero.

Fit and reconcile

(b) The two correlation matrices:

	Pearson			
	x_1	x_2	x_3	x_4
x_1	1.0000	0.7991	0.7195	-0.0047
x_2	0.7991	1.0000	0.5876	-0.0209
x_3	0.7195	0.5876	1.0000	-0.0086
x_4	-0.0047	-0.0209	-0.0086	1.0000

	Spearman			
	x_1	x_2	x_3	x_4
x_1	1.0000	0.9719	0.6839	0.0002
x_2	0.9719	1.0000	0.6674	-0.0011
x_3	0.6839	0.6674	1.0000	-0.0024
x_4	0.0002	-0.0011	-0.0024	1.0000

The largest gap is exactly x_1 vs. x_2 : Pearson = 0.7991, Spearman = 0.9719, gap is **0.1728**.

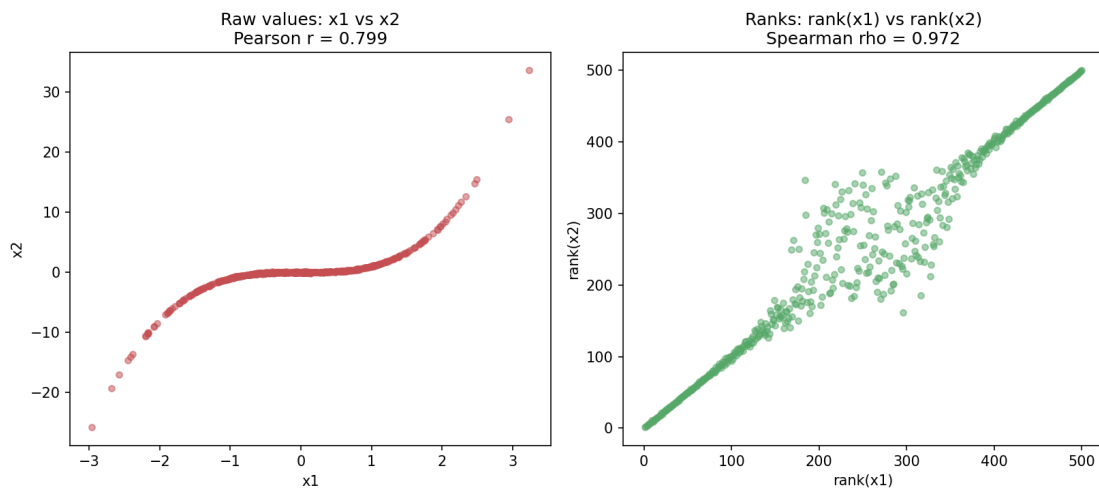


Figure 5: Left: raw x_1 vs. x_2 . Right: same data on rank axes.

(c) x_2 is a monotonically increasing but non-linear function of x_1 : order is preserved almost everywhere, but the size of the change is not proportional, growing faster in the tails than the middle. Pearson can only fit a straight line to that curve, so it reports a value pulled down toward 0.80. Spearman works on ranks, so it only cares about order, and reports ≈ 0.97 . **Spearman is a better measure of dependence here**; Pearson indicates how well a straight line fits.

4 Problem 4

Predict

Sample means and covariance matrix of (x_1, x_2) :

$\text{mean}(x_1)$	$\text{mean}(x_2)$	$\text{var}(x_1)$	$\text{var}(x_2)$	$\text{cov}(x_1, x_2)$	$\text{corr}(x_1, x_2)$
-0.0167	0.0453	1.0999	4.1047	1.6969	0.7986

(a) Week 2, Section 2.4 gives $\bar{\Sigma} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$ for the block we condition on. We want $x_2 | x_1$, so block 1 (unknown) is x_2 and block 2 (known) is x_1 . Then

$$\text{var}(x_2 | x_1) = \text{var}(x_2) - \frac{\text{cov}(x_1, x_2)^2}{\text{var}(x_1)} = 4.1047 - \frac{1.6969^2}{1.0999} = \mathbf{1.4867}.$$

The reduction factor is $\text{var}(x_2 | x_1) / \text{var}(x_2) = 1 - \rho^2 = \mathbf{0.3622}$: conditioning on x_1 shrinks our uncertainty about x_2 to about 36% of its unconditional variance.

(b) No. The term $\Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$ in the conditional variance formula contains no observed value a . It is unlike the conditional *mean* formula, which depends on a through $(a - \mu_2)$. Under joint normality, conditioning on x_1 always shrinks the variance by the same factor, no matter what value of x_1 we see.

Fit and reconcile

(c) The conditional mean is $E[x_2 | x_1 = a] = \mu_2 + (\text{cov}(x_1, x_2) / \text{var}(x_1))(a - \mu_1) = 0.0711 + 1.5428a$. The coefficient 1.5428 is exactly the OLS slope of x_2 on x_1 (Week 2, Section 3.2), confirmed by a direct OLS regression.

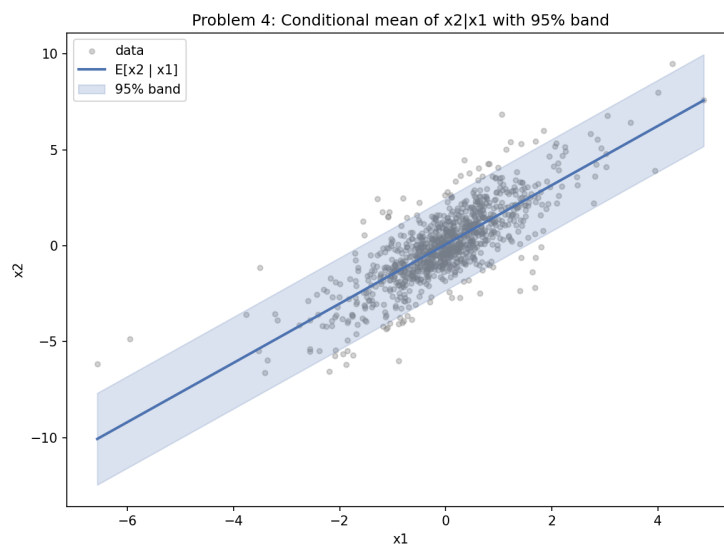


Figure 6: x_2 vs. x_1 with the conditional mean and a constant-width 95% band.

(d) Overall coverage of the band is **93.50%**, below the nominal 95%.

(e) Coverage by distance from $\text{mean}(x_1)$:

Bucket	n	Coverage
Within 1 sd	759	95.65%
1–2 sd	190	85.79%
Beyond 2 sd	51	90.20%

(f) Coverage is not flat: near nominal at the center, well below it in both outer buckets. Residual standard deviation by bucket rises from 1.076 to 1.580 to 1.596, so the true conditional variance **grows with** $|x_1 - \mu_1|$. Which means the data are heteroskedastic. This is exactly the assumption from part (b) failing: $\bar{\Sigma}$ is not really constant, so one band width is too narrow in the tails. The conditional **mean** in part (c) still survives as the best linear predictor. what fails is using a single $\bar{\Sigma}$ to describe uncertainty everywhere.

5 Problem 5

Predict

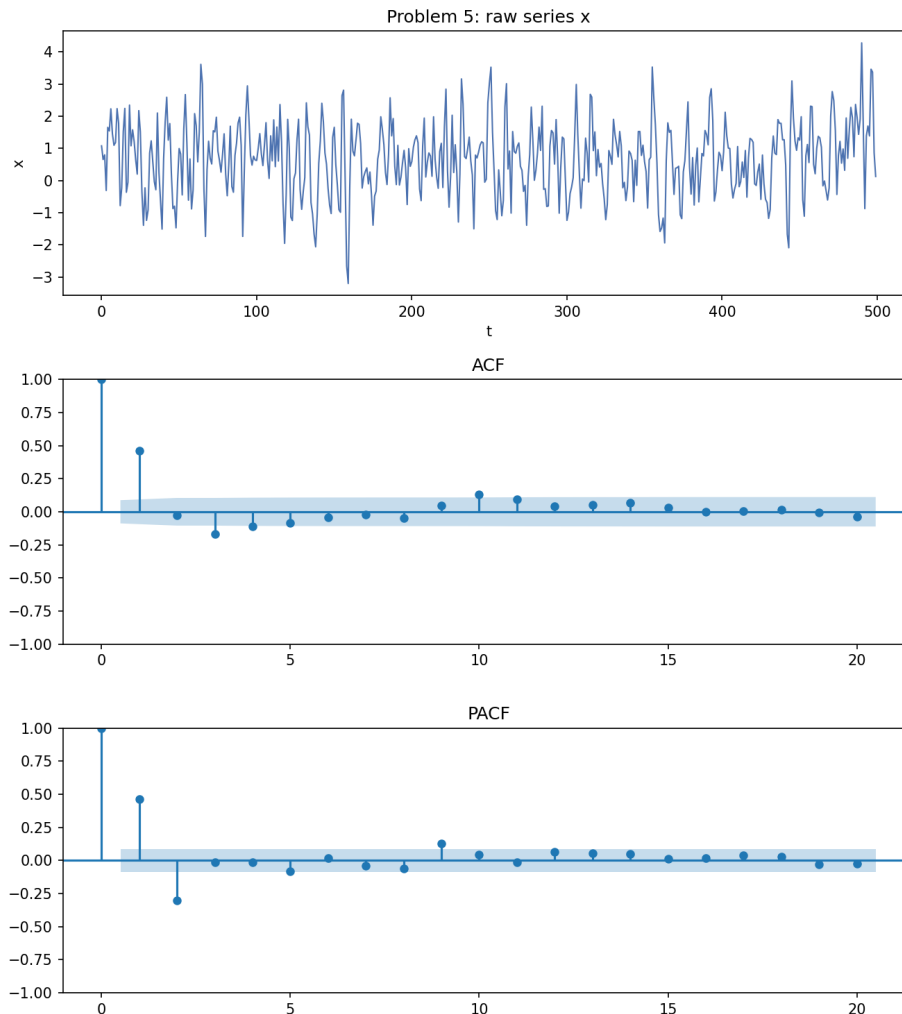


Figure 7: Series x , its ACF, and its PACF (20 lags).

(a) The **PACF cuts off** (significant at lags 1 and 2 only: 0.4616, -0.3031) and the **ACF decays** in an oscillating way. By Week 2, Table 3 (AR(p): ACF decays, PACF cuts off after lag p), this is an **AR(2)** process.

(b) A lag counts as significant if it falls outside $\pm 1.96/\sqrt{n} = \pm 0.0877$ (Week 2, Section 6.1). “Cuts off” means all later lags stay inside that band, allowing for the occasional chance crossing expected 1 in 20 times at the 5% level (e.g. the isolated PACF spike at lag 9).

Fit and reconcile

(c) AICc for each model, with $k = p + d$, $p = (\text{order}) + 1$, $d = 1$ (Week 2, Section 5.3):

	AR(1)	AR(2)	AR(3)	MA(1)	MA(2)	MA(3)
AICc	1418.59	1372.44	1374.34	1389.08	1381.23	1374.12

(d) AICc selects **AR(2)**, matching the prediction in part (a).

(e) AR(2) coefficients are 0.6012, -0.3026 ; AR(3) adds a third coefficient of only -0.0165 , improving the log-likelihood by just 0.07 while costing one extra parameter, so AICc *increases* ($1372.44 \rightarrow 1374.34$) and rejects AR(3). R^2 barely moves ($0.28573 \rightarrow 0.28593$) but can only increase as parameters are added (Week 2, Section 5.1), so it has no way to penalize the extra, non-explanatory coefficient the way AICc does.